

Ellipsoid-Fit Magnetometer Calibration: Complete Derivation of the Estimation Algorithm

Starting point. The calibration model $\mathbf{m} = A\mathbf{h} + \mathbf{b}$ with $\|\mathbf{h}\| = H$ and $M \triangleq A^{-\top}A^{-1}$ implies that every raw measurement lies on the ellipsoid

$$(\mathbf{m} - \mathbf{b})^\top M (\mathbf{m} - \mathbf{b}) = H^2. \quad (1)$$

This document derives, with every algebraic step shown, how N raw measurements $\mathbf{m}_1, \dots, \mathbf{m}_N$ are converted into estimates of A , $\mathbf{b}$, and H .

Notation. $\mathbf{m} \in \mathbb{R}^3$ measured vector, $\mathbf{h} \in \mathbb{R}^3$ true field, $A \in \mathbb{R}^{3 \times 3}$ calibration matrix, $\mathbf{b} \in \mathbb{R}^3$ hard-iron bias, $H = \|\mathbf{h}\|$, $M = A^{-\top}A^{-1} \in \mathbb{R}^{3 \times 3}$. Components of a generic measurement are written $\mathbf{m} = [x \ y \ z]^\top$.

Two structural properties of M are used repeatedly:

$$M^\top = \left(A^{-\top}A^{-1}\right)^\top = A^{-\top}A^{-1} = M \quad (M \text{ is symmetric}), \quad (2)$$

$$\mathbf{x}^\top M \mathbf{x} = \mathbf{x}^\top A^{-\top}A^{-1}\mathbf{x} = (A^{-1}\mathbf{x})^\top (A^{-1}\mathbf{x}) = \|A^{-1}\mathbf{x}\|^2 > 0 \quad \text{for } \mathbf{x} \neq \mathbf{0}, \quad (3)$$

where (3) uses invertibility of A ($A^{-1}\mathbf{x} \neq \mathbf{0}$ when $\mathbf{x} \neq \mathbf{0}$). Hence $M \succ 0$: symmetric positive definite.

1 Expansion into the general quadratic equation

Expand the left side of (1) term by term:

$$(\mathbf{m} - \mathbf{b})^\top M (\mathbf{m} - \mathbf{b}) = \mathbf{m}^\top M \mathbf{m} - \mathbf{m}^\top M \mathbf{b} - \mathbf{b}^\top M \mathbf{m} + \mathbf{b}^\top M \mathbf{b}. \quad (4)$$

The middle two terms are scalars, and a scalar equals its own transpose:

$$\mathbf{m}^\top M \mathbf{b} = \left(\mathbf{m}^\top M \mathbf{b}\right)^\top = \mathbf{b}^\top M^\top \mathbf{m} \stackrel{(2)}{=} \mathbf{b}^\top M \mathbf{m}. \quad (5)$$

Substituting (5) into (4) and equating to H^2 :

$$\mathbf{m}^\top M \mathbf{m} - 2\mathbf{b}^\top M \mathbf{m} + \mathbf{b}^\top M \mathbf{b} - H^2 = 0. \quad (6)$$

Now write (6) in components. Since M is symmetric, it has the form

$$M = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{12} & M_{22} & M_{23} \\ M_{13} & M_{23} & M_{33} \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (7)$$

With $\mathbf{m} = [x \ y \ z]^\top$, the quadratic term is

$$\mathbf{m}^\top M \mathbf{m} = M_{11}x^2 + M_{22}y^2 + M_{33}z^2 + 2M_{12}xy + 2M_{13}xz + 2M_{23}yz, \quad (8)$$

where the factors of 2 arise because each off-diagonal entry appears twice in the double sum $\sum_i \sum_j M_{ij} m_i m_j$ (once as $M_{ij} m_i m_j$ and once as $M_{ji} m_j m_i = M_{ij} m_i m_j$).

Define the auxiliary linear-coefficient vector and constant

$$\mathbf{u} \triangleq -M\mathbf{b} = [u_1 \ u_2 \ u_3]^\top \in \mathbb{R}^3, \quad (9)$$

$$s \triangleq \mathbf{b}^\top M\mathbf{b} - H^2 \in \mathbb{R}. \quad (10)$$

Then $-2\mathbf{b}^\top M\mathbf{m} = 2\mathbf{u}^\top \mathbf{m} = 2u_1x + 2u_2y + 2u_3z$, and (6) becomes the *general quadric equation*

$$M_{11}x^2 + M_{22}y^2 + M_{33}z^2 + 2M_{12}xy + 2M_{13}xz + 2M_{23}yz + 2u_1x + 2u_2y + 2u_3z + s = 0. \quad (11)$$

Intuition. The ellipsoid (1) is written in “center–shape” form, which is nonlinear in the physical parameters ($\mathbf{b}$ appears inside a quadratic form with M). Equation (11) trades those physical parameters for *algebraic* coefficients $\{M_{ij}, u_k, s\}$ in which the equation is *linear*. This is the entire trick: fit the linear algebraic form first, undo the substitutions (9)–(10) afterwards.

2 Why exactly ten unknown coefficients

Count the independent coefficients in (11):

- Quadratic part: $M \in \mathbb{R}^{3 \times 3}$ symmetric $\Rightarrow \frac{3(3+1)}{2} = 6$ independent entries: $M_{11}, M_{22}, M_{33}, M_{12}, M_{13}, M_{23}$.
- Linear part: $\mathbf{u} \in \mathbb{R}^3 \Rightarrow 3$ entries: u_1, u_2, u_3 .
- Constant part: $s \in \mathbb{R} \Rightarrow 1$ entry.

Total: $6 + 3 + 1 = 10$. Collect them in the parameter vector

$$\boldsymbol{\theta} \triangleq [M_{11}, M_{22}, M_{33}, M_{12}, M_{13}, M_{23}, u_1, u_2, u_3, s]^\top \in \mathbb{R}^{10}. \quad (12)$$

Remark 1 (Only 9 degrees of freedom). Equation (11) is homogeneous in $\boldsymbol{\theta}$: if $\boldsymbol{\theta}$ satisfies it for every point on the surface, so does $\kappa\boldsymbol{\theta}$ for any $\kappa \neq 0$, because multiplying every coefficient by κ multiplies the left side by κ and $\kappa \cdot 0 = 0$. The surface therefore determines $\boldsymbol{\theta}$ only up to scale: 10 coefficients, 9 degrees of freedom. This single fact drives Sections 6–8 (the trivial-solution problem and its cure) and Section 11 (the unobservability of H).

3 Linear equation for a single measurement

Take one raw measurement $\mathbf{m}_k = [x_k \ y_k \ z_k]^\top$. If the sensor were noiseless, $\mathbf{m}_k$ would satisfy (11) exactly. Group the measurement-dependent factors into a row vector and the unknowns into $\boldsymbol{\theta}$:

$$\mathbf{d}(\mathbf{m}_k) \triangleq [x_k^2, y_k^2, z_k^2, 2x_ky_k, 2x_kz_k, 2y_kz_k, 2x_k, 2y_k, 2z_k, 1] \in \mathbb{R}^{1 \times 10}. \quad (13)$$

Multiplying (13) by (12) reproduces the left side of (11) term by term:

$$\begin{aligned} \mathbf{d}(\mathbf{m}_k) \boldsymbol{\theta} &= x_k^2 M_{11} + y_k^2 M_{22} + z_k^2 M_{33} + 2x_ky_k M_{12} + 2x_kz_k M_{13} + 2y_kz_k M_{23} \\ &\quad + 2x_ku_1 + 2y_ku_2 + 2z_ku_3 + s. \end{aligned} \quad (14)$$

Hence each measurement contributes one scalar linear equation in the 10 unknowns:

$$\boxed{\mathbf{d}(\mathbf{m}_k) \boldsymbol{\theta} = 0, \quad k = 1, \dots, N.} \quad (15)$$

Intuition. The quadratic nonlinearity has been pushed entirely into the *known* data (the squares and cross-products in (13) are computed from numbers you already have). What remains unknown enters *linearly*. Each measurement is one linear constraint on $\boldsymbol{\theta}$; ten unknowns with one scale freedom means a minimum of 9 well-placed measurements, and thousands of measurements give a heavily overdetermined system averaged over noise.

4 Stacking N measurements

Stack the N rows (13) into the *design matrix*

$$D \triangleq \begin{bmatrix} \mathbf{d}(\mathbf{m}_1) \\ \mathbf{d}(\mathbf{m}_2) \\ \vdots \\ \mathbf{d}(\mathbf{m}_N) \end{bmatrix} = \begin{bmatrix} x_1^2 & y_1^2 & z_1^2 & 2x_1y_1 & 2x_1z_1 & 2y_1z_1 & 2x_1 & 2y_1 & 2z_1 & 1 \\ x_2^2 & y_2^2 & z_2^2 & 2x_2y_2 & 2x_2z_2 & 2y_2z_2 & 2x_2 & 2y_2 & 2z_2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_N^2 & y_N^2 & z_N^2 & 2x_Ny_N & 2x_Nz_N & 2y_Nz_N & 2x_N & 2y_N & 2z_N & 1 \end{bmatrix} \in \mathbb{R}^{N \times 10}. \quad (16)$$

The N scalar equations (15) become one matrix equation:

$$D\boldsymbol{\theta} = \mathbf{0}_N. \quad (17)$$

With sensor noise, no $\boldsymbol{\theta}$ satisfies (17) exactly; instead define the residual vector

$$\mathbf{r}(\boldsymbol{\theta}) \triangleq D\boldsymbol{\theta} \in \mathbb{R}^N, \quad r_k = \mathbf{d}(\mathbf{m}_k)\boldsymbol{\theta}, \quad (18)$$

whose k -th entry measures how far measurement k lies from the quadric defined by $\boldsymbol{\theta}$ (in the *algebraic* sense — the value of the quadric polynomial at $\mathbf{m}_k$, not the Euclidean point-to-surface distance).

5 The least-squares cost function

Penalize the residuals by their squared Euclidean norm:

$$J(\boldsymbol{\theta}) \triangleq \|\mathbf{r}(\boldsymbol{\theta})\|^2 = \sum_{k=1}^N (\mathbf{d}(\mathbf{m}_k)\boldsymbol{\theta})^2. \quad (19)$$

Expand using (18):

$$J(\boldsymbol{\theta}) = (D\boldsymbol{\theta})^\top (D\boldsymbol{\theta}) = \boldsymbol{\theta}^\top D^\top D \boldsymbol{\theta} = \boldsymbol{\theta}^\top S \boldsymbol{\theta}, \quad S \triangleq D^\top D \in \mathbb{R}^{10 \times 10}. \quad (20)$$

S is the *scatter matrix* of the problem. Two properties follow directly:

$$S^\top = (D^\top D)^\top = D^\top D = S \quad (\text{symmetric}), \quad (21)$$

$$\boldsymbol{\theta}^\top S \boldsymbol{\theta} = \|D\boldsymbol{\theta}\|^2 \geq 0 \quad \forall \boldsymbol{\theta} \quad (\text{positive semidefinite}). \quad (22)$$

Intuition. $J(\boldsymbol{\theta})$ is the total squared algebraic error of the fitted quadric over all data. Minimizing an “algebraic” rather than geometric distance is what keeps the problem solvable in closed form; the cost is a mild, data-dependent bias (Remark 4).

6 Why ordinary least squares collapses to the trivial solution

Consider the unconstrained problem $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$. The gradient of the quadratic form (20) is

$$\nabla_{\boldsymbol{\theta}} J = \nabla_{\boldsymbol{\theta}} (\boldsymbol{\theta}^\top S \boldsymbol{\theta}) = (S + S^\top) \boldsymbol{\theta} \stackrel{(21)}{=} 2S\boldsymbol{\theta}. \quad (23)$$

Setting (23) to zero gives the normal equations $S\boldsymbol{\theta} = \mathbf{0}$, which always admit

$$\boldsymbol{\theta} = \mathbf{0}, \quad J(\mathbf{0}) = 0. \quad (24)$$

Since $J \geq 0$ by (22), $\boldsymbol{\theta} = \mathbf{0}$ attains the *global* minimum. Any unconstrained solver returns it (or, with noisy data and full-rank S , returns it uniquely). But $\boldsymbol{\theta} = \mathbf{0}$ means all ten quadric coefficients vanish — “the equation $0 = 0$ ” — which describes no surface at all.

The root cause is Remark 1: because the model (15) is homogeneous with a zero right-hand side, the cost satisfies

$$J(\kappa\boldsymbol{\theta}) = (\kappa\boldsymbol{\theta})^\top S(\kappa\boldsymbol{\theta}) = \kappa^2 J(\boldsymbol{\theta}), \quad (25)$$

so shrinking any candidate ($\kappa \rightarrow 0$) always reduces the cost. There is no data-driven mechanism to fix the overall scale of $\boldsymbol{\theta}$; the scale must be pinned by an explicit *constraint*.

7 The constrained least-squares problem

Fix the scale with a quadratic normalization constraint. Let $C \in \mathbb{R}^{10 \times 10}$ be a chosen symmetric constraint matrix and require

$$\boxed{\min_{\boldsymbol{\theta} \in \mathbb{R}^{10}} \boldsymbol{\theta}^\top S \boldsymbol{\theta} \quad \text{subject to} \quad \boldsymbol{\theta}^\top C \boldsymbol{\theta} = 1.} \quad (26)$$

The simplest and most common choice is

$$C = I_{10} \quad \Longleftrightarrow \quad \|\boldsymbol{\theta}\|^2 = 1, \quad (27)$$

which forbids $\boldsymbol{\theta} = \mathbf{0}$ while treating all directions of $\boldsymbol{\theta}$ equally. (An alternative choice of C that additionally *guarantees* the fitted quadric is an ellipsoid is discussed in Remark 2.)

Form the Lagrangian with multiplier $\lambda \in \mathbb{R}$:

$$\mathcal{L}(\boldsymbol{\theta}, \lambda) = \boldsymbol{\theta}^\top S \boldsymbol{\theta} - \lambda (\boldsymbol{\theta}^\top C \boldsymbol{\theta} - 1). \quad (28)$$

Stationarity in $\boldsymbol{\theta}$ (using (23) for both quadratic forms, valid because S and C are symmetric):

$$\nabla_{\boldsymbol{\theta}} \mathcal{L} = 2S\boldsymbol{\theta} - 2\lambda C\boldsymbol{\theta} = \mathbf{0}. \quad (29)$$

Stationarity in λ recovers the constraint:

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(\boldsymbol{\theta}^\top C \boldsymbol{\theta} - 1) = 0 \quad \Longrightarrow \quad \boldsymbol{\theta}^\top C \boldsymbol{\theta} = 1. \quad (30)$$

8 The generalized eigenvalue problem

Dividing (29) by 2:

$$\boxed{S\boldsymbol{\theta} = \lambda C\boldsymbol{\theta}.} \quad (31)$$

This is a *generalized eigenvalue problem* (GEP) in the pair (S, C) : the candidate solutions of (26) are exactly the generalized eigenvectors of (S, C) , scaled to satisfy $\boldsymbol{\theta}^\top C \boldsymbol{\theta} = 1$.

Which eigenpair minimizes the cost? Left-multiply (31) by $\boldsymbol{\theta}^\top$ and use the constraint:

$$J(\boldsymbol{\theta}) = \boldsymbol{\theta}^\top S \boldsymbol{\theta} = \lambda \boldsymbol{\theta}^\top C \boldsymbol{\theta} \stackrel{(30)}{=} \lambda. \quad (32)$$

The cost at any stationary point equals its eigenvalue. Therefore:

$$\hat{\boldsymbol{\theta}} = \text{generalized eigenvector of } (S, C) \text{ with the smallest admissible eigenvalue } \lambda. \quad (33)$$

For $C = I_{10}$, (31) reduces to the ordinary symmetric eigenvalue problem $S\boldsymbol{\theta} = \lambda\boldsymbol{\theta}$; since $S = D^\top D$, the minimizer is equivalently the right singular vector of D associated with its smallest singular value — numerically, one SVD of D solves the whole fit:

$$D = U\Sigma V^\top, \quad \hat{\boldsymbol{\theta}} = V[:, 10] \text{ (last column, smallest singular value)}. \quad (34)$$

Intuition. The data build a 10-dimensional quadratic “cost bowl” $\boldsymbol{\theta}^\top S \boldsymbol{\theta}$. The constraint surface $\boldsymbol{\theta}^\top C \boldsymbol{\theta} = 1$ is a shell around the origin. The minimum of the bowl restricted to the shell occurs along the direction where the bowl rises most slowly — the eigenvector of smallest eigenvalue. With perfect noiseless data lying exactly on one ellipsoid, $D\boldsymbol{\theta} = 0$ has an exact nontrivial solution, S is rank 9, $\lambda_{\min} = 0$, and the fit is exact.

Remark 2 (Ellipsoid-specific constraint). $C = I_{10}$ does not by itself guarantee the minimizing quadric is an ellipsoid rather than another quadric type (hyperboloid, paraboloid); with magnetometer data that sweeps a reasonable range of orientations, the ellipsoid is what fits and the issue rarely bites, and positive definiteness is checked in Section 9 anyway. If a guarantee is required, choose C to encode an ellipsoid-only normalization (Li & Griffiths, 2004): with $I_M \triangleq M_{11} + M_{22} + M_{33}$ and J_M the sum of the 2×2 principal minors of M , the constraint $4J_M - I_M^2 = 1$ is quadratic in the first six entries of $\boldsymbol{\theta}$ and hence expressible as $\boldsymbol{\theta}^\top C \boldsymbol{\theta} = 1$ with a constant indefinite C ; (31) then must genuinely be solved as a GEP, keeping the eigenvector whose eigenvalue is positive and whose M is definite.

9 From fitted coefficients to $\mathbf{b}$ and M

Unpack the estimate $\hat{\boldsymbol{\theta}} = [\hat{\theta}_1, \dots, \hat{\theta}_{10}]^\top$ into the quadric’s matrix form:

$$Q \triangleq \begin{bmatrix} \hat{\theta}_1 & \hat{\theta}_4 & \hat{\theta}_5 \\ \hat{\theta}_4 & \hat{\theta}_2 & \hat{\theta}_6 \\ \hat{\theta}_5 & \hat{\theta}_6 & \hat{\theta}_3 \end{bmatrix} \in \mathbb{R}^{3 \times 3}, \quad \hat{\mathbf{u}} \triangleq \begin{bmatrix} \hat{\theta}_7 \\ \hat{\theta}_8 \\ \hat{\theta}_9 \end{bmatrix} \in \mathbb{R}^3, \quad \hat{s} \triangleq \hat{\theta}_{10} \in \mathbb{R}. \quad (35)$$

Because the fit determines $\boldsymbol{\theta}$ only up to the unknown scale κ (Remark 1), the correspondence with the physical quantities in (9)–(10) is

$$Q = \kappa M, \quad \hat{\mathbf{u}} = -\kappa M \mathbf{b}, \quad \hat{s} = \kappa (\mathbf{b}^\top M \mathbf{b} - H^2), \quad \kappa \neq 0 \text{ unknown}. \quad (36)$$

Sign fix. $M \succ 0$ by (3), but the eigen-solver may return $-\hat{\boldsymbol{\theta}}$ (eigenvectors have arbitrary sign), making Q negative definite. Test Q ’s eigenvalues; if they are all negative, replace $\hat{\boldsymbol{\theta}} \leftarrow -\hat{\boldsymbol{\theta}}$ (equivalently $Q \leftarrow -Q$, $\hat{\mathbf{u}} \leftarrow -\hat{\mathbf{u}}$, $\hat{s} \leftarrow -\hat{s}$). After this, $\kappa > 0$ and $Q \succ 0$. (Mixed signs mean the fit is not an ellipsoid — insufficient orientation coverage; collect better data or use Remark 2.)

Hard-iron bias $\mathbf{b}$

From the second identity in (36), $\hat{\mathbf{u}} = -\kappa M \mathbf{b} = -Q \mathbf{b}$, hence

$$\boxed{\hat{\mathbf{b}} = -Q^{-1} \hat{\mathbf{u}}}. \quad (37)$$

Verification that κ cancels:

$$-Q^{-1} \hat{\mathbf{u}} = -(\kappa M)^{-1} (-\kappa M \mathbf{b}) = \kappa^{-1} M^{-1} \kappa M \mathbf{b} = \mathbf{b}. \quad (38)$$

The bias is fully observable — it is the geometric *center* of the ellipsoid, which is scale-invariant.

The scaled shape matrix M/H^2

Eliminate κ using the constant term. From the third identity in (36),

$$\kappa H^2 = \kappa \mathbf{b}^\top M \mathbf{b} - \hat{s}. \quad (39)$$

Express $\kappa \mathbf{b}^\top M \mathbf{b}$ in fitted quantities: substituting $\mathbf{b} = -Q^{-1}\hat{\mathbf{u}}$ from (37) and $\kappa M = Q$,

$$\kappa \mathbf{b}^\top M \mathbf{b} = \mathbf{b}^\top (\kappa M) \mathbf{b} = \mathbf{b}^\top Q \mathbf{b} = (-Q^{-1}\hat{\mathbf{u}})^\top Q (-Q^{-1}\hat{\mathbf{u}}) = \hat{\mathbf{u}}^\top Q^{-\top} Q Q^{-1} \hat{\mathbf{u}} = \hat{\mathbf{u}}^\top Q^{-1} \hat{\mathbf{u}}, \quad (40)$$

where $Q^{-\top} = Q^{-1}$ because Q is symmetric. Substituting (40) into (39):

$$\kappa H^2 = \hat{\mathbf{u}}^\top Q^{-1} \hat{\mathbf{u}} - \hat{s}. \quad (41)$$

Now divide $Q = \kappa M$ by the scalar (41):

$$\boxed{\widetilde{M} \triangleq \frac{Q}{\hat{\mathbf{u}}^\top Q^{-1} \hat{\mathbf{u}} - \hat{s}} = \frac{\kappa M}{\kappa H^2} = \frac{M}{H^2}.} \quad (42)$$

The unknown κ cancels exactly, but what survives is the *ratio* M/H^2 , not M itself. Consistency check: with $\widetilde{M} = M/H^2$ the ellipsoid (1) reads $(\mathbf{m} - \mathbf{b})^\top \widetilde{M} (\mathbf{m} - \mathbf{b}) = 1$, i.e., $\widetilde{M}$ is the shape matrix of the ellipsoid normalized to “radius” 1 — precisely what the data determine.

Intuition. The data cloud is a geometric surface in $\mathbb{R}^3$. A surface pins down its own center ($\mathbf{b}$: 3 numbers) and its shape-with-size ($\widetilde{M}$: 6 numbers) — 9 quantities, matching the 9 degrees of freedom of Remark 1. The 10th quantity, the split of $\widetilde{M} = M/H^2$ into M and H separately, is invisible to the fit. That is Section 11.

10 Recovering A from $M = A^{-\top} A^{-1}$

Suppose a value of H has been chosen (Section 11), so that

$$\widehat{M} = H^2 \widetilde{M} \quad (43)$$

is the estimate of M itself, symmetric positive definite. Invert the defining relation:

$$M = A^{-\top} A^{-1} \implies M^{-1} = (A^{-\top} A^{-1})^{-1} = (A^{-1})^{-1} (A^{-\top})^{-1} = A A^\top, \quad (44)$$

using $(XY)^{-1} = Y^{-1} X^{-1}$ and $(A^{-\top})^{-1} = A^\top$. So A is any matrix “square-root factor” of M^{-1} .

Route 1: symmetric square root via eigendecomposition

$\widehat{M} \succ 0$ admits the spectral decomposition

$$\widehat{M} = V \Lambda V^\top, \quad V \in \mathbb{R}^{3 \times 3} \text{ orthogonal } (V^\top V = I_3), \quad \Lambda = \text{diag}(\lambda_1, \lambda_2, \lambda_3), \quad \lambda_i > 0. \quad (45)$$

Then

$$\widehat{M}^{-1} = (V \Lambda V^\top)^{-1} = V^{-\top} \Lambda^{-1} V^{-1} = V \Lambda^{-1} V^\top, \quad (46)$$

because $V^{-1} = V^\top$. Define

$$\boxed{\hat{A} \triangleq V \Lambda^{-1/2} V^\top, \quad \Lambda^{-1/2} = \text{diag}(\lambda_1^{-1/2}, \lambda_2^{-1/2}, \lambda_3^{-1/2}).} \quad (47)$$

Verification of (44): $\hat{A}$ is symmetric ($\hat{A}^\top = V \Lambda^{-1/2} V^\top = \hat{A}$), so

$$\hat{A} \hat{A}^\top = V \Lambda^{-1/2} V^\top V \Lambda^{-1/2} V^\top = V \Lambda^{-1/2} \Lambda^{-1/2} V^\top = V \Lambda^{-1} V^\top = \widehat{M}^{-1}. \checkmark \quad (48)$$

Direct verification of the original definition: $\hat{A}^{-1} = V \Lambda^{1/2} V^\top$ (invert the middle factor), and

$$\hat{A}^{-\top} \hat{A}^{-1} = (V \Lambda^{1/2} V^\top)^\top V \Lambda^{1/2} V^\top = V \Lambda^{1/2} V^\top V \Lambda^{1/2} V^\top = V \Lambda V^\top = \widehat{M}. \checkmark \quad (49)$$

Route 2: Cholesky factorization

Compute the Cholesky factorization of the inverse,

$$\widehat{M}^{-1} = L L^T, \quad L \in \mathbb{R}^{3 \times 3} \text{ lower triangular, positive diagonal,} \quad (50)$$

and set $\hat{A} = L$. Then $\hat{A}\hat{A}^T = LL^T = \widehat{M}^{-1}$ by construction, and equivalently

$$\hat{A}^{-T}\hat{A}^{-1} = L^{-T}L^{-1} = (LL^T)^{-1} = \widehat{M}. \checkmark \quad (51)$$

(In practice, factor $\widehat{M} = GG^T$ and take $L = G^{-T}$, avoiding an explicit matrix inverse.)

Remark 3 (A is only determined up to a rotation). *If $\hat{A}$ satisfies (44), so does $\hat{A}R$ for any orthogonal $R \in \mathbb{R}^{3 \times 3}$:*

$$(\hat{A}R)(\hat{A}R)^T = \hat{A}RR^T\hat{A}^T = \hat{A}\hat{A}^T = \widehat{M}^{-1}. \quad (52)$$

This is not a defect of the algorithm; it is physics. The data record only field magnitudes through the ellipsoid; a fixed rotation of $\mathbf{h}$ before it enters the sensor is invisible. Routes 1 and 2 give different (equally valid) representatives; the symmetric root (47) is the standard convention. If the true soft-iron distortion contains a rotational part, pure magnetometer data cannot recover it — an external reference (accelerometer, known heading) is required to fix R .

11 Recovering H — and why it must be chosen

Section 9 showed the data determine only the ratio $\widetilde{M} = M/H^2$ (42). The reason is visible in the model itself: for any $\alpha > 0$, the substitution

$$A \rightarrow \alpha A, \quad \mathbf{h} \rightarrow \alpha^{-1} \mathbf{h} \quad (\text{so } H \rightarrow \alpha^{-1} H) \quad (53)$$

leaves $\mathbf{m} = A\mathbf{h} + \mathbf{b}$ unchanged while scaling $M = A^{-T}A^{-1} \rightarrow \alpha^{-2}M$, i.e., M/H^2 is invariant. **H is structurally unobservable from magnetometer data alone.** It must be fixed by convention or by external information:

- (a) **Normalization (recommended when only orientation matters).** Choose $H = 1$. Then $\widehat{M} = \widetilde{M}$ in (43), and calibrated measurements have unit norm — exactly what quaternion/AHRS measurement updates consume, since they use only the field *direction*.
- (b) **Physical units.** Set $H = H_{\text{ref}}$, the local geomagnetic field magnitude from a reference model (IGRF/WMM) for your location and date. Then $\widehat{M} = H_{\text{ref}}^2 \widetilde{M}$ and $\hat{A}$ carries the correct physical scale, so $\hat{A}^{-1}(\mathbf{m} - \hat{\mathbf{b}})$ is in true field units.

Under either choice, H propagates through (43)–(47) as a simple scalar: comparing (47) applied to $H^2 \widetilde{M}$ versus $\widetilde{M}$,

$$\hat{A}(H) = H^{-1} \hat{A}(1), \quad \hat{A}(H)^{-1} = H \hat{A}(1)^{-1}, \quad (54)$$

so changing H after the fact rescales the calibrated vectors without changing their directions.

12 Proof that calibration maps the data onto a sphere

Proposition 1. *Let $\hat{A}$, $\hat{\mathbf{b}}$, $\widehat{M}$ satisfy $\hat{A}^{-T}\hat{A}^{-1} = \widehat{M}$ exactly (Routes 1 or 2 both do). For any point $\mathbf{m}$ lying on the fitted ellipsoid $(\mathbf{m} - \hat{\mathbf{b}})^T \widehat{M} (\mathbf{m} - \hat{\mathbf{b}}) = H^2$, the calibrated measurement*

$$\mathbf{m}_{\text{cal}} \triangleq \hat{A}^{-1} (\mathbf{m} - \hat{\mathbf{b}}) \quad (55)$$

satisfies $\|\mathbf{m}_{\text{cal}}\| = H$, i.e., lies on the sphere of radius H centered at the origin.

Proof. Compute the squared norm directly from (55):

$$\|\mathbf{m}_{\text{cal}}\|^2 = \mathbf{m}_{\text{cal}}^\top \mathbf{m}_{\text{cal}} \quad (56)$$

$$= \left(\hat{A}^{-1}(\mathbf{m} - \hat{\mathbf{b}}) \right)^\top \left(\hat{A}^{-1}(\mathbf{m} - \hat{\mathbf{b}}) \right) \quad (57)$$

$$= (\mathbf{m} - \hat{\mathbf{b}})^\top \hat{A}^{-\top} \hat{A}^{-1} (\mathbf{m} - \hat{\mathbf{b}}) \quad (58)$$

$$= (\mathbf{m} - \hat{\mathbf{b}})^\top \widehat{M} (\mathbf{m} - \hat{\mathbf{b}}) \quad (59)$$

$$= H^2, \quad (60)$$

where (58) uses $(XY)^\top = Y^\top X^\top$, (59) substitutes the defining relation $\hat{A}^{-\top} \hat{A}^{-1} = \widehat{M}$, and (60) holds because $\mathbf{m}$ lies on the fitted ellipsoid. Taking the positive square root gives $\|\mathbf{m}_{\text{cal}}\| = H$. $\square$

For real (noisy) data, points lie *near* the fitted ellipsoid, so $\|\mathbf{m}_{\text{cal}}\| = H + \varepsilon$ with ε the residual noise; the spread of $\|\mathbf{m}_{\text{cal}}\|$ around H is the standard post-calibration quality metric.

Intuition. $\hat{A}^{-1}$ undoes the soft-iron distortion (it whitens the quadratic form: substituting $\mathbf{m} - \hat{\mathbf{b}} = \hat{A} \mathbf{w}$ turns the ellipsoid equation into $\mathbf{w}^\top \mathbf{w} = H^2$), and subtracting $\hat{\mathbf{b}}$ undoes the hard-iron offset. The rotational ambiguity of Remark 3 is invisible here: a sphere maps to itself under rotation, which is exactly *why* R is unobservable.

13 Complete algorithm

Ellipsoid-fit magnetometer calibration

Input: raw measurements $\mathbf{m}_k = [x_k \ y_k \ z_k]^\top$, $k = 1, \dots, N$ ($N \gg 10$), covering diverse orientations.

Output: $\hat{A}$, $\hat{\mathbf{b}}$, H , calibrated data $\{\mathbf{m}_{\text{cal},k}\}$.

1. **Design matrix.** Build $D \in \mathbb{R}^{N \times 10}$ row-by-row from (13)–(16). (*Numerics: pre-center and pre-scale the data — subtract the mean, divide by the mean radius — to condition D ; undo the transform on $\hat{\mathbf{b}}, \widetilde{M}$ afterwards.*)
2. **Fit.** Compute the SVD $D = U\Sigma V^\top$ and set $\hat{\boldsymbol{\theta}} = V[:, 10]$ (34) (equivalently: eigenvector of $S = D^\top D$ with smallest eigenvalue; or solve the GEP (31) with the Li–Griffiths C of Remark 2 for a guaranteed ellipsoid).
3. **Unpack.** Form Q , $\hat{\mathbf{u}}$, $\hat{s}$ from $\hat{\boldsymbol{\theta}}$ via (35).
4. **Sign fix.** If Q ’s eigenvalues are all negative, negate Q , $\hat{\mathbf{u}}$, $\hat{s}$. If mixed signs: abort — data do not define an ellipsoid; recollect with better orientation coverage.
5. **Bias.** $\hat{\mathbf{b}} = -Q^{-1}\hat{\mathbf{u}}$ (37).
6. **Normalized shape.** $\widetilde{M} = Q/(\hat{\mathbf{u}}^\top Q^{-1}\hat{\mathbf{u}} - \hat{s})$ (42). Verify $\widetilde{M} \succ 0$.
7. **Choose H .** $H = 1$ (unit-norm output) or $H = H_{\text{ref}}$ from IGRF/WMM. Set $\widehat{M} = H^2 \widetilde{M}$ (43).
8. **Recover A .** Eigendecompose $\widehat{M} = V\Lambda V^\top$ and set $\hat{A} = V\Lambda^{-1/2}V^\top$ (47) (or Cholesky, (50)). Note $\hat{A}$ is one representative of the family $\hat{A}R$ (52).
9. **Calibrate.** $\mathbf{m}_{\text{cal},k} = \hat{A}^{-1}(\mathbf{m}_k - \hat{\mathbf{b}})$ for all k (55). (Compute via a linear solve or the closed form $\hat{A}^{-1} = V\Lambda^{1/2}V^\top$; do not form an explicit inverse in embedded code.)
10. **Validate.** Check that $\|\mathbf{m}_{\text{cal},k}\|$ clusters tightly around H (e.g., relative standard deviation of the norms). Large or orientation-correlated residuals indicate uncompensated disturbances, not fit failure.

Remark 4 (Scope of optimality). *The estimator above minimizes algebraic error (19), not geometric distance or a likelihood; under measurement noise it is consistent in practice but not exactly maximum-likelihood, and its residual weighting depends on where points sit on the ellipsoid. For the last few percent of accuracy, use $\hat{A}, \hat{\mathbf{b}}$ as the initialization for a nonlinear ML refinement (e.g., Gauss–Newton on the geometric residuals, or the joint calibration formulations in Kok–Hol–Schön). For consumer-grade MEMS magnetometers, the algebraic fit alone is normally sufficient.*